

PRICE: A Certified Exchange-Rate Framework for Pricing Self-Knowledge in Performative Markets

1st Shriraghav Ashok
Haas School of Business
University of California, Berkeley
Berkeley, United States
shriraghav.a524@gmail.com

Abstract—Every trading desk that quotes prices retrains its models on the trades that come back, but the quotes themselves decide which trades arrive. Retraining on such data settles at a quote that is stable but not the most profitable one, and the shortfall is paid on every trade: in a public corporate-bond calibration it is 1.9–4.3% of the quoted spread. Closing it requires knowing how the desk’s own quotes move the flow, learnable only by quoting something other than what looks best, at a cost. Existing methods treat that cost as zero; this paper prices it. We show that along any steady exploration policy the variance of the estimated response times the cumulative cost of exploring equals $\frac{1}{2}\gamma_{PO}\sigma^2$, a constant of the market; that knowing the functional form of the response buys nothing within two curvature lengths of the quote; and that the cheapest exploration follows the flat directions of the desk’s own objective. We confirm every closed form on a simulated dealer market (36 preregistered checks, 0 failures), a second economy, and a public corporate-bond calibration reproduced in 10 of 10 cells. We then build SafeD-PerfGD, an agent that explores on the cheapest support and corrects its quotes only when a certificate guarantees the retraining loop stays stable; over 12 seeds no baseline beats it on both median final error and median regret. Ultimately, PRICE lets a desk read the price of self-knowledge off calibrated constants: how much it costs, how little one can pay, where to spend, and whether to spend at all.

I. INTRODUCTION

A. Problem Definition

A market maker’s profit is driven by the order flow its quotes attract, and the same quotes decide what the desk’s models see next. Tighter quotes win more volume but attract more informed (toxic) flow, trades from counterparties who know more than the dealer, so the quoted spread shapes the very data the desk later learns from [1], [2]. Retraining on data the deployment produced is a fixed-point iteration, the setting of performative prediction [3], [4], and it converges to a stable point that is not the performative optimum [5]. In the corporate-bond calibration of REFLEX [6] the gap between the two is 1.9–4.3% of the quoted spread across ten cells. The problem is acute now that quoting models are retrained on live order flow [2], [6]. The gap is paid on every trade for as long as the loop runs, and the common assumption behind closing it is that the exploration needed is free. Closing it requires an estimate of how deployments move

the distribution, obtainable only by deploying something other than what looks best, at a real cost in the desk’s own objective.

B. Current Approaches

Looking at the existing literature, Perdomo et al. [3] showed that repeated risk minimization converges when the distribution’s sensitivity to the deployment is small relative to the curvature of the loss, and Miller et al. [5] separated the stable point from the optimum. Algorithms that reach the optimum run from performative gradient descent [7] to its descendants [8], [9], [10], [11]; Jagadeesan et al. [12] bound the regret of finding the optimum, and Lin and Zrnic [13] and Bracale et al. [14] fit structural models of the map. However, these methods estimate the response by perturbing deployments without pricing the perturbations in the objective being optimized.

Additionally, operations research and economics have long studied a seller learning demand while pricing [15], [16], [17], [18], [19], [20], where the $\sqrt{T}$ regret lower bounds [21], [22] rest on the device that information is bounded by price dispersion, which costs regret. Dual control [23], [24], task-optimal exploration [25], [26], and optimal design [27], [28], [29] prescribe what to probe. However, none of these approaches prices exploration inside a retraining loop whose own stability constrains it: the demand curve or plant is exogenous, and the tradeoff is a proof device rather than an instrument a desk can compute.

This blind spot is particularly consequential now that retraining loops run on live order flow, since a desk has no calibrated instrument for how much self-knowledge to buy, where to buy it, or whether the resulting correction keeps the loop stable.

C. Proposed Approach

To address these limitations, this paper proposes a three-stage framework. First, we derive an exchange-rate identity that fixes the product of estimation variance and cumulative performative regret at $\frac{1}{2}\gamma_{PO}\sigma^2$ along any stationary exploration, and minimax floors showing that no policy, adaptive or not, pays less by more than a constant, structure-proof within an exact reach $2/c$ of the operating point. Second, we derive the exploration shape that minimizes risk at a budget priced by the desk’s own curvature, $\Gamma_{PO}^{-1/2}$, and the factor by which isotropic exploration [30] overpays. Third, we build SafeD-PerfGD, an agent that explores on the optimal support and corrects the deployment only when a perturbed-modulus certificate guarantees that the retraining loop keeps contracting, and verify every closed form on the live pipeline.

D. Results Overview

The framework's closed forms hold on the live pipeline to within Monte Carlo error. The exchange-rate product is measured at 0.01062 and 0.01070 against a predicted 0.01056, isotropic exploration overpays the optimal design by 1.529 against a predicted 1.506, the within-reach Cramér–Rao ratio bottoms out at 1.0005, and the port reproduces the published REFLEX run in 10 of 10 cells. SafeD-PerfGD reaches the optimum with zero trust-region violations, and over 12 seeds no baseline Pareto-dominates it on median final error and regret, while the unsafe gradient baseline pays more than $5 \times$ its median regret. These results reveal that the price of self-knowledge is a property of the market rather than of the explorer: where a desk probes, not how much, is what it controls.

II. BACKGROUND

A. The Dealer Market and the Retraining Loop

This paper builds upon the operational setting of a dealer who quotes a half-spread h and refits that quote to the flow it observes. Informed (toxic) flow arrives at rate $\tau(h) = C_0 + C_1 e^{-ch}$, so tighter quotes summon more of it; profit $J(h; \tau)$ at a frozen toxic level is concave in h with curvature $\gamma > 0$, each unit of toxic flow being worth $(h - \psi)$ for adverse severity ψ , and the performative value of a quote is $\Phi(h) = J(h, \tau(h))$. Blind retraining on the observed flow, the current practice, contracts if and only if the modulus $m = \varepsilon/\gamma$ is below one, converging to the stable point $h^* = h_{\text{SP}}$, the anchor throughout, while the optimum $h_{\text{PO}} = \arg \max \Phi$ sits at a tighter spread. The curvature $\gamma_{\text{PO}} = -\Phi''(h^*)$ of the performative objective at the anchor prices what follows.

B. Target Quantity – The Response Slope and Its Price

The quantity a desk must learn to leave the anchor is the response slope $\varepsilon(h) = -\tau'(h) > 0$, how much more toxic flow a unit of tightening attracts. With deviation $d_t = h_t - h^*$, exploration input u_t , and noisy observed flow, the linearized loop and observation are

$$d_{t+1} = -md_t + u_t, \quad \tau_t = \tau(h^*) - \varepsilon d_t + \sigma \zeta_t \quad (1)$$

In (1), d_t = deviation from the anchor, m = the retraining modulus, u_t = the deliberate exploration step, $\sigma \zeta_t$ = response noise. Every deviation that teaches the desk about ε is itself a trade at a worse price, a structure we term priced observation. Exploration is paid in one of two currencies, the deviation budget $D_T = \sum_{t \leq T} d_t^2$ or the value budget $C_T = \sum_{t \leq T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$, the realized cost in the dealer's own objective. The estimator $\hat{\varepsilon}$ is the least-squares slope of flow on spread with the sign flipped.

C. Optimal Design and Certified Correction

This paper additionally builds upon two tools. For where to probe, it builds upon optimal experimental design [27], [28] and the van Trees inequality [29], which bounds every estimator under an adaptive design. For when to act, it builds upon anytime-valid confidence sequences and a perturbed-modulus argument that turns a confidence set for the response into a certificate on the closed-loop modulus. The certificate is required

because a correction is deployable only if the loop it modifies provably keeps contracting.

III. METHODS

A. Data Collection

All data comes from the public REFLEX calibration and from simulated markets whose constants are set by it [6]. The real-data leg uses ten corporate-bond cells, investment-grade and high-yield ratings crossed with five regimes, over 170 CUSIPs with at least 12 months of returns. For each cell the port recovers the stable half-spread h^* , the response slope ε^* , and the modulus m , and extends them to γ_{PO} , the exchange rate $\frac{1}{2}\gamma_{\text{PO}}$, and the gap $h_{\text{SP}} - h_{\text{PO}}$. Only the benign-flow channel, the noise scale, and the spread are data-identified; the toxic channel is structurally scaled. The mechanism-level leg uses the simulated dealer market of Section III-C and a linear-quadratic pricing economy whose deployed price anchors demand with gain g .

B. The Exchange-Rate Identity and Minimax Floors

Under smoothness and symmetric exploration the value budget is the design energy priced at the curvature, $C_T = \frac{1}{2}\gamma_{\text{PO}}\mathbb{E}[D_T] + R_T$, with a remainder of third order in the deviations. For the least-squares slope on any stationary exploration policy, conditional on the design and on every realization,

$$\text{Var}(\hat{\varepsilon}) \times \frac{1}{2}\gamma_{\text{PO}}D_T = \frac{1}{2}\gamma_{\text{PO}}\sigma^2(1 + O_P(1/T)) \quad (2)$$

and nothing about the policy or the modulus appears. The mechanism is cancellation: information and cost both scale with design energy, so their ratio is a property of the market. Even the $O(\log T)$ schedules of Lai and Robbins [31] sit on the frontier.

Under a deviation budget D , over all non-anticipating policies and all estimators, the minimax risk is at least $\sigma^2/(D + 4\pi^2\sigma^2/w^2)$ for ε in an interval of width w . The product with the budget cost is therefore at least $\frac{1}{2}\gamma_{\text{PO}}\sigma^2(1 - o(1))$, and symmetric probing attains it, so this floor is exact. Under a value budget measured against the certainty-equivalent deployment, the product is at least $\frac{1}{27}\gamma_{\text{PO}}\sigma^2(1 - O(T^{-1/2}))$. This floor lies within a factor 13.5 of the achievable product.

C. The Simulated Dealer Market

We simulate the dealer of Section II-A on a live deploy–observe–fit–correct pipeline at moduli $m \in \{0.3, 0.6, 0.85\}$ for the saturation checks and at $m \in \{0.3, 0.6\}$ with exploration amplitudes $s_e \in \{0.05, 0.15\}$ for the exchange-rate cells, and run the agent at a horizon of $T = 4000$ and the ablation grid at $T = 4500$. Six assumptions carry the theory: the performative objective is three times differentiable and strongly concave near the anchor (A1); policies are non-anticipating, symmetric for the identities but unrestricted for the adaptive floors; the response noise is independent standard Gaussian, used only for the floors; deviations stay inside a trust region $|d_t| \leq r$ (A4); design energy grows linearly in T ; and the modulus is below one. Two choices are critical to validity. First, the cost in (2) along the loop (1) is the realized loss in the dealer's own objective against the jitter-free loop at the anchor, the currency a desk pays. Second, one cell deliberately leaves the scope of (A1), a saturating environ-

ment in which the objective flattens on the wide side of the anchor; its drift is reported, not excused.

D. Exploration Design and Estimation

In d dimensions with the curvature Γ_{PO} of Φ pricing the budget $\text{tr}(\Gamma_{\text{PO}}W) \leq B$ on the exploration second moment W , the A-optimal design, which minimizes $\text{tr}W^{-1}$, is $W^* = B\Gamma_{\text{PO}}^{-1/2} / \text{tr}\Gamma_{\text{PO}}^{1/2}$; the D-optimal design is $(B/d)\Gamma_{\text{PO}}^{-1}$; and the c-optimal design for a correction direction q is $Bqq^\top / (q^\top \Gamma_{\text{PO}} q)$. Isotropic exploration at the same budget overpays the A-optimal design by exactly

$$F = d \text{tr}(\Gamma_{\text{PO}}) / \left(\text{tr}\Gamma_{\text{PO}}^{1/2} \right)^2 \in [1, d] \quad (3)$$

with $F = 1$ if and only if all curvatures are equal, so a desk should explore where its objective is flat, to the square-root degree. The agent explores on the D-optimal three-point support re-centered at the operating point and clipped to the trust region $r \leq 0.8$, inside the reach $2/c$ (1.333 here) within which the floor is structure-proof; it fits the family (C_0, C_1, c) by profile least squares, carries an anytime confidence width ci_t for $\hat{\epsilon}$, and observes only its own realized flow. Against the secant estimator, which shares its variance at budget B but carries an $O(B/(\gamma_{\text{PO}}T))^2$ bias from the third derivative, a structural fit with misspecification δ_{mis} wins if and only if $\delta_{\text{mis}} < |\tau'''(h^*)|B/(3\gamma_{\text{PO}}T)$, at short horizons and never asymptotically in T [13].

E. The SafeD-PerfGD Agent and Evaluation

As summarized in Fig. 1, SafeD-PerfGD composes the three results per deployment round. It explores on the D-optimal support inside the trust region; fits (C_0, C_1, c) with the confidence width ci_t ; applies the correction $-(h - \psi)\hat{\epsilon}$ with Newton scaling $\eta = 1/\hat{\gamma}_{\text{PO}}$ only if $\eta \text{ci}_t L_{\text{fam}} \leq c_{\text{gate}}$, where $L_{\text{fam}} = 1 + \hat{c}|h^* - \psi|$ is the family's Lipschitz constant and c_{gate} is chosen so that the worst-case modulus stays below $1 - c_{\text{margin}}$; and otherwise freezes the correction while the blind step, stable at $m < 1$, keeps running. Evaluation follows a preregistered harness of 36 measured-versus-predicted rows. We compare the agent with finite-difference PerfGD [7], zeroth-order P&L optimization, a performative-confidence-bounds explorer [12], and blind retraining over 12 seeds; ablate the design, anchor, and gate; transplant the theory unchanged to the linear-quadratic pricing economy; and instantiate it on the REFLEX calibration.

IV. RESULTS AND ANALYSIS

A. External Validation of the Simulated Market

Before evaluating the price, we validate the market it is read from. As shown in Fig. 2, the REFLEX port reproduces the published stable half-spread h^* , response slope ϵ^* , and modulus m in 10 of 10 cells, with a maximum deviation of 0.23% in h^* , 0.62% in ϵ^* , and 0.76% in m , across anchors spanning 0.435 (IG-calm) to 4.167 (HY-crisis) and slopes spanning 465.4 to 2.2. Because the agreement holds in every cell across two orders of magnitude in the quantities the theory prices, the port reproduces the published calibration rather than matching a single point, and every subsequent result on the real-data leg inherits

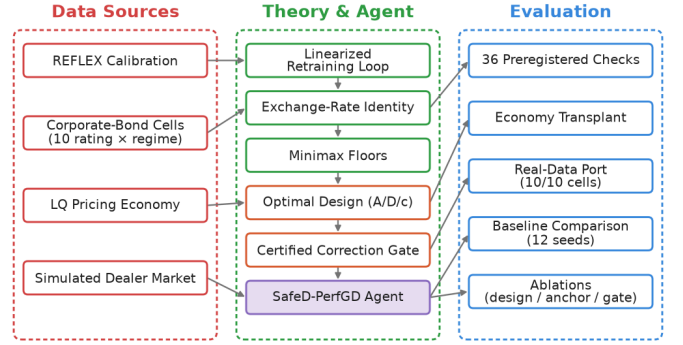


Fig. 1. Pipeline architecture for the proposed pricing framework and certified agent followed by evaluation and benchmarking

this grounding. On the validated cells the exchange rate $\frac{1}{2}\gamma_{\text{PO}}$ spans 255.3 to 1.2, and the gap $h_{\text{SP}} - h_{\text{PO}}$ is 1.9–4.3% of the quoted spread.

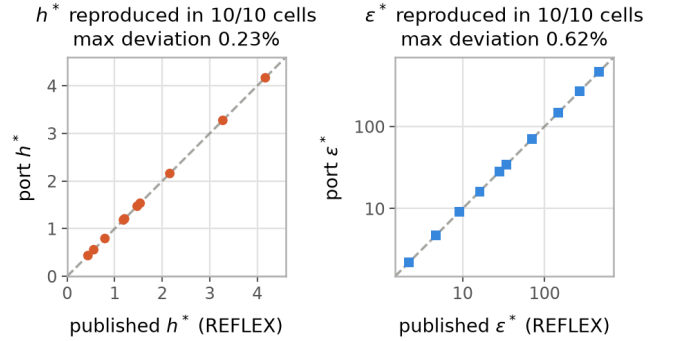


Fig. 2. External validation of the simulated market against the published REFLEX calibration in 10 rating-and-regime cells

B. The Exchange Rate: Global Verification

As shown in Fig. 3 and TABLE I, the measured product $\text{Var}(\hat{\epsilon}) \times C_T$ sits on the predicted constant in every in-scope cell, at 0.01062 and 0.01070 at exploration amplitudes 0.05 and 0.15 against a predicted $\frac{1}{2}\gamma_{\text{PO}}\sigma^2 = 0.01056$ at modulus 0.3, and 0.00549 and 0.00569 against 0.00535 at modulus 0.6. Tripling the amplitude moves the product by under 1% at the low modulus and about 3.5% at the high one, inside the preregistered 15% tolerance. The saturating cell, outside the strong-concavity scope, drifts to 0.0062244, 16.3% above the constant, so the 36 rows close as 35 passes and one out-of-scope drift cell. Tripling the amplitude at fixed modulus and doubling the modulus at fixed amplitude together establish the paper's central dissociation: the amplitude of exploration changes how fast the dealer learns and how much it pays, but not the exchange rate between them.

C. Where the Price Can Be Lowered: Shape, Not Amount

Because the unit price is fixed, the operational question is where to explore, not how much. As shown in the left panel of Fig. 4, isotropic exploration in eight dimensions overpays the A-optimal shape by a measured factor 1.529 against the predicted $F = 1.506$ of (3). On the REFLEX calibration the dispersion factor is $F = 1.63$ across the ten-cell portfolio against 1.002

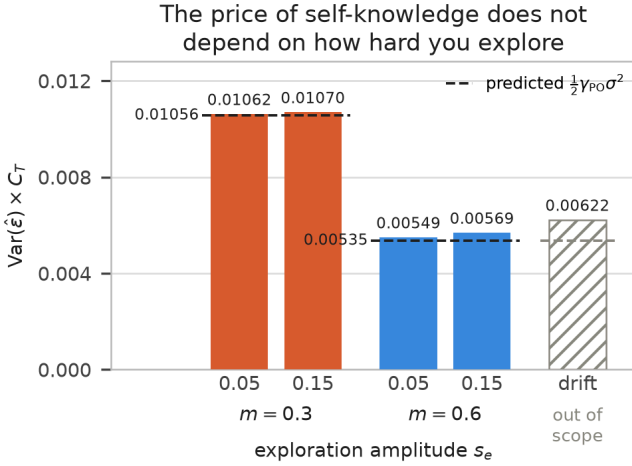


Fig. 3. Exchange-rate invariance across exploration amplitudes and moduli, with the out-of-scope drift cell shown for completeness

TABLE I

SELECTED PREREGISTERED VERIFICATION ROWS (36 ROWS, 0 FAILURES)

Claim	Measured	Predicted
Exchange rate, $m = 0.3$	0.01062, 0.01070	0.01056
Exchange rate, $m = 0.6$	0.00549, 0.00569	0.00535
Isotropic/A-optimal ratio, $d = 8$	1.529	$F = 1.506$
Within-reach min Cramér–Rao ratio	1.0005	≥ 1
Beyond-reach witness ratio	0.8517	< 1
SafeD settles at h_{PO}	1.258	1.296
Trust-region violations	0	0
LQ pricing exchange rate, $g = 0.4, 0.7$	0.384, 0.192	0.384, 0.192

within one book. As shown in the right panel of Fig. 4, a policy that knows the response family might shop for designs whose parametric Cramér–Rao product undercuts the floor, but restricted to the reach $h \geq h^* - 2/c$ a hardened search over such designs bottoms out at a ratio of 1.0005. Adapting the amplitude does not help either, since constant and decaying schedules evaluated at the true cost land at 1.018 and 0.995, within Monte Carlo error of the floor. Beyond the reach the floor breaks. A four-point design with one probe of weight 3×10^{-4} below $h^* - 2/c$ attains 0.8517.

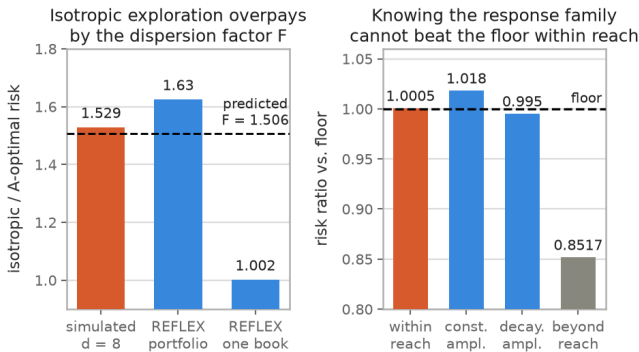


Fig. 4. Where the price can be lowered: the dispersion factor paid by isotropic exploration, and the exact reach within which neither knowing the response family nor adapting the exploration amplitude beats the floor

The exchange rate also prices whether to explore at all. Exploring once to variance v and correcting forever, discounted at rate ρ_{disc} , has net present value $(A - av)/\rho_{\text{disc}} - b/v$ with $A = \frac{1}{2}\gamma_{PO}(h_{SP} - h_{PO})^2$, $a = \frac{1}{2}\gamma_{PO}\kappa^2$, $b = \frac{1}{2}\gamma_{PO}\sigma^2$, and $\kappa = |dh_{PO}/d\varepsilon|$. It is maximized at $v^* = (\sigma/\kappa)\sqrt{\rho_{\text{disc}}}$ and positive if and only if $\rho_{\text{disc}} < (h_{SP} - h_{PO})^4/(4\kappa^2\sigma^2)$. The curvature cancels, so whether to buy self-knowledge depends on the gap, the noise, and the patience of the desk, not on how steep its objective is.

D. Certified Correction

The corrected map replaces the true response by an estimate with error $e = \hat{\varepsilon} - \varepsilon$, and its slope at the fixed point differs from the ideal modulus $\rho = 1 - \eta\gamma_{PO}$ by at most $\eta\|e\|_{\infty} + \eta\|h^* - \psi\|e'\|_{\infty}$. The pessimism rule corrects only when the worst-case modulus over an anytime-valid confidence set at level $1 - \alpha$ respects the margin, and so keeps the realized modulus below $1 - c_{\text{margin}}$ at every step with probability at least $1 - \alpha$. On the live pipeline SafeD-PerfGD settles at $h = 1.258$ against the performative optimum $h_{PO} = 1.296$, a 3.0% gap inside the preregistered 8% band. The trust region is never violated, and the largest step taken is 0.88 of the cap. In the agent verification run the gate clears after roughly 1,300 deployments, a freeze fraction of 0.33 of its $T = 4000$ horizon, during which the blind step keeps the loop converging. The gate opens only after design energy accumulates, a regret payment by (2) that Section IV-E makes visible. The certificate, not the point estimate, is the deployable object.

E. Baseline Comparison

As shown in Fig. 5 and TABLE II, over 12 seeds no baseline Pareto-dominates SafeD-PerfGD on median final error and regret. Finite-difference PerfGD [7] attains a lower median final error of 0.0263 against 0.0599, but pays a median regret of 9586.79 against 1007.86, more than $5 \times$ as much, because its unsafe perturbations are never priced. The UCB-Grid explorer [12] attains the lowest median regret at 315.92 but a final error of 0.1547 against 0.0599, and it edges SafeD-PerfGD in one of the twelve seeds, a cost of certification we report deliberately. Zeroth-order optimization attains a marginally lower median regret of 952.33 at a final error of 0.3868 against 0.0599, and blind retraining trails on both at 1.2293 and 2115.54.

TABLE II
BASELINES OVER 12 SEEDS (MEDIAN)

Method	Final error	Regret
SafeD-PerfGD	0.0599	1007.86
FD-PerfGD	0.0263	9586.79
ZO-PerfOpt	0.3868	952.33
UCB-Grid	0.1547	315.92
BlindRRM	1.2293	2115.54

F. Ablation Analysis

Ablations. Switching the design, anchor, and gate off in every combination, Fig. 6 reveals that the anchor carries the accuracy and the gate carries the safety. With the anchor on, the final error is 0.0437 with the D-optimal design and 0.0182 with full-support jitter, whether or not the gate is active; with the anchor off, it rises to 0.1845 (gated) and 0.1216 (ungated) under the

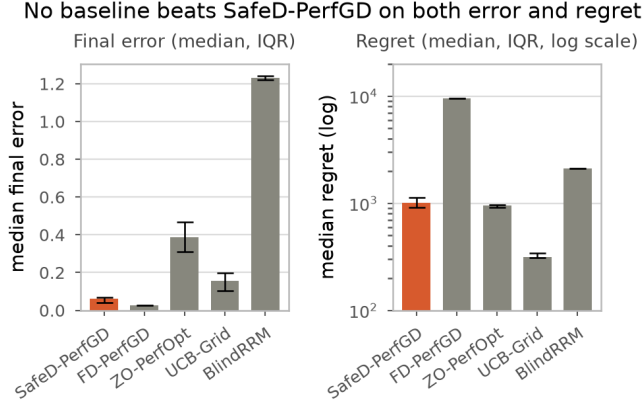


Fig. 5. Baseline comparison over 12 seeds on median final error and median regret with interquartile ranges

design and to 0.2825 (gated) and 0.0939 (ungated) under jitter. The anchor-off cells limit-cycle, so every final error is a tail average. The gate’s effect is on discipline, not accuracy. In the ablation grid, run at $T = 4500$, gated cells freeze for a fraction 0.554 of the run under the design with or without the anchor, 0.338 under jitter with the anchor, and 0.680 under jitter with neither anchor nor design, while ungated cells never freeze; the 0.554 there and the 0.33 of the agent verification run come from two different runs. The design-off cells still converge at matched transit energy, so the design’s advantage is allocational, not a cliff. What the design buys is identification of the curvature c . In the agent verification run the design arm’s $\text{sd}(\hat{c})$ of 0.23364 matches the Fisher prediction 0.24784, while the jitter arm’s is 7.8 times larger, leaving c unidentified.

Robustness. As shown in Fig. 7, the theory transplants unchanged to a second economy and across the modulus. In the linear-quadratic pricing economy the exchange rate is exact in all four cells, 0.384 at gain $g = 0.4$ and 0.192 at $g = 0.7$, at both exploration amplitudes 0.2 and 0.5, because concavity is global in the linear-quadratic case and no cell saturates; the transplanted agent reaches the pricing optimum at 12.65 against 12.5. The information a loop yields with no exploration at all

was also confirmed across the modulus. With no probing the design energy saturates at $d_0^2/(1 - m^2)$, so $\text{Var}(\hat{\epsilon}) \geq \sigma^2(1 - m^2)/d_0^2$ for any unbiased estimator, and blind retraining buys this free information with an $O(1/T)$ bias that changes sign at $m = 1/3$. No blind trajectory identifies the family’s curvature c : information saturates for $m < 1$, and the two-point orbit at $m = 1$ cannot identify three parameters, which is why the design arm’s identification of c above is bought, not free. Across moduli $m = 0.3, 0.6$, and 0.85 the information-saturation cap holds at 0.010857, 0.014994, and 0.033748 against 0.010989, 0.015625, and 0.036036, within 6.3% at the most reflexive loop. The matching feedback floor, observation noise re-entering as deployment excitation of variance $(\sigma/\gamma)^2/(1 - m^2)$, holds at the same three moduli within 0.2–6.4%.

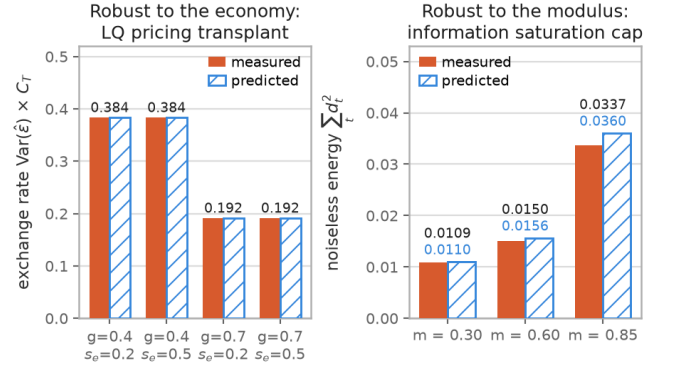


Fig. 7. Robustness of the exchange rate to the economy and of the saturation cap to the modulus (labels rounded)

Taken together, the results form a single chain: a validated market (Fig. 2) establishes that the product of slope variance and realized cost is a constant of the market, unchanged by how hard a desk explores (Fig. 3, TABLE I); the shape of exploration therefore becomes the binding decision (Fig. 4); the certified agent is the only one competitive on both error and regret (Fig. 5, TABLE II), with accuracy traceable to the anchor and safety to the gate (Fig. 6); and the identity holds across the economy and the modulus (Fig. 7).

V. CONCLUSION

The proposed framework prices a decision that current retraining methods treat as free, and the price holds to within Monte Carlo error. The product of estimation variance and cumulative performative regret is measured at 0.01062 and 0.01070 against a predicted 0.01056, isotropic exploration overpays the optimal shape by 1.529 against a predicted 1.506, and 36 preregistered rows close with 0 failures. The certified agent reaches the optimum with zero trust-region violations, and no baseline Pareto-dominates it over 12 seeds.

The verification confirms the mechanism: information and cost both scale with the design energy a desk spends, so their ratio, $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$, is a property of the market rather than of the explorer. No policy pays much less, knowing the response family cannot help within two curvature lengths of the quote, and the cheapest exploration follows $\Gamma_{\text{PO}}^{-1/2}$, the flat directions of the desk’s own objective.

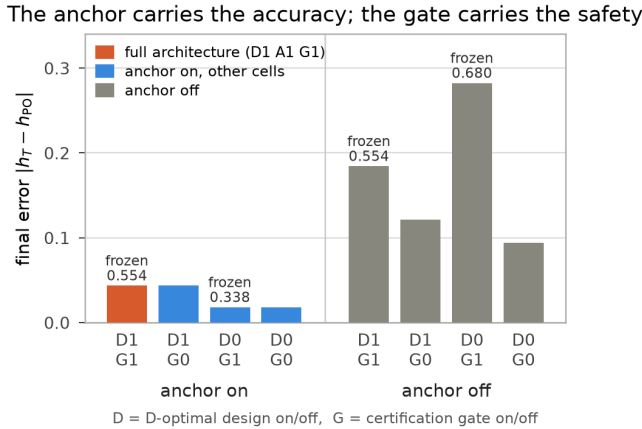


Fig. 6. Ablation of the design, anchor, and gate components on final error (300-step tail average of $|h_t - h_{\text{PO}}|$), with freeze fractions annotated for gated cells

Ultimately, the findings suggest that a retraining loop on live order flow need not treat self-knowledge as free: a desk can read off calibrated constants how much it costs to learn its own influence, how little it can pay, where to pay it, and whether the resulting correction keeps the loop stable.

VI. LIMITATIONS AND FUTURE WORK

While the framework achieves close agreement with its closed forms, certain constraints and assumptions merit discussion.

The identity is local. It holds to leading order in the exploration amplitude, and the saturating cell drifts 16.3% above the constant; the trust region is therefore load-bearing. Beyond the reach $2/c$ the floor provably fails (the 0.8517 witness), and at finite amplitude wide-side probing rides the objective's flattening to a ratio of 0.966 under the true cost. The value-budget floor is proved to within a factor 13.5 of the achievable product; the sharp constant $1/2$ is numerically supported, not proved. Certification is conditional on the validity of the confidence sequence, and a conjectured $O(\sqrt{T})$ identification-regret bound has a proof strategy only.

Additionally, the real-data leg validates the machinery rather than the effect size. The toxic channel is structurally scaled, and the port is validated against REFLEX's published run, not trade-level data.

Future work will close the sharp constant and extend the floors to multi-dealer markets. It will also make certification survive misspecification of the response family and test the predicted long-horizon reversal on live order-flow data.

REFERENCES

- [1] O. Guéant, C.-A. Lehalle, and J. Fernandez-Tapia, "Dealing with the Inventory Risk: A Solution to the Market Making Problem," *arXiv preprint arXiv:1105.3115*, 2013.
- [2] A. Barzykin, P. Bergault, O. Guéant, and M. Lemmel, "Optimal Quoting under Adverse Selection and Price Reading," *arXiv preprint arXiv:2508.20225*, 2026.
- [3] J. C. Perdomo, T. Zrnic, C. Mendler-Dünner, and M. Hardt, "Performative Prediction," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- [4] M. Hardt and C. Mendler-Dünner, "Performative Prediction: Past and Future," *arXiv preprint arXiv:2310.16608*, 2023.
- [5] J. Miller, J. C. Perdomo, and T. Zrnic, "Outside the Echo Chamber: Optimizing the Performative Risk," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- [6] V. Nagarajan and S. Ashok, "REFLEX: Reflexive Equilibrium Fixed-point Learning for Endogenous eXchanges," *arXiv preprint arXiv:2608.16155*, 2026.
- [7] Z. Izzo, L. Ying, and J. Zou, "How to Learn when Data Reacts to Your Model: Performative Gradient Descent," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- [8] C. Mendler-Dünner, J. C. Perdomo, T. Zrnic, and M. Hardt, "Stochastic Optimization for Performative Prediction," in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020.
- [9] D. Drusvyatskiy and L. Xiao, "Stochastic Optimization with Decision-Dependent Distributions," *Mathematics of Operations Research*, vol. 48, no. 2, pp. 954–998, 2023.
- [10] G. Brown, S. Hod, and I. Kalemaj, "Performative Prediction in a Stateful World," in *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- [11] Q. Li and H.-T. Wai, "State Dependent Performative Prediction with Stochastic Approximation," in *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- [12] M. Jagadeesan, T. Zrnic, and C. Mendler-Dünner, "Regret Minimization with Performative Feedback," in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.
- [13] L. Lin and T. Zrnic, "Plug-in Performative Optimization," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [14] D. Bracale, S. Maity, Y. Sun, and M. Banerjee, "Learning the Distribution Map in Reverse Causal Performative Prediction," in *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, in PMLR, vol. 258, 2025.
- [15] M. Rothschild, "A Two-Armed Bandit Theory of Market Pricing," *Journal of Economic Theory*, vol. 9, no. 2, pp. 185–202, 1974.
- [16] J. M. Harrison, N. B. Keskin, and A. Zeevi, "Bayesian Dynamic Pricing Policies: Learning and Earning Under a Binary Prior Distribution," *Management Science*, vol. 58, no. 3, pp. 570–586, 2012.
- [17] A. V. den Boer, "Dynamic Pricing and Learning: Historical Origins, Current Research, and New Directions," *Surveys in Operations Research and Management Science*, vol. 20, no. 1, pp. 1–18, 2015.
- [18] D. Easley and N. M. Kiefer, "Controlling a Stochastic Process with Unknown Parameters," *Econometrica*, vol. 56, no. 5, pp. 1045–1064, 1988.
- [19] P. Aghion, P. Bolton, C. Harris, and B. Jullien, "Optimal Learning by Experimentation," *The Review of Economic Studies*, vol. 58, no. 4, pp. 621–654, 1991.
- [20] S. J. Grossman and J. E. Stiglitz, "On the Impossibility of Informationally Efficient Markets," *The American Economic Review*, vol. 70, no. 3, pp. 393–408, 1980.
- [21] J. Broder and P. Rusmevichientong, "Dynamic Pricing Under a General Parametric Choice Model," *Operations Research*, vol. 60, no. 4, pp. 965–980, 2012.
- [22] N. B. Keskin and A. Zeevi, "Dynamic Pricing with an Unknown Demand Model: Asymptotically Optimal Semi-Myopic Policies," *Operations Research*, vol. 62, no. 5, pp. 1142–1167, 2014.
- [23] A. A. Feldbaum, "Dual Control Theory I," *Automation and Remote Control*, vol. 21, pp. 874–880, 1960.
- [24] X. Bombois, G. Scorletti, M. Gevers, P. M. J. Van den Hof, and R. Hildebrand, "Least Costly Identification Experiment for Control," *Automatica*, vol. 42, no. 10, pp. 1651–1662, 2006.
- [25] A. Wagenmaker, M. Simchowitz, and K. Jamieson, "Task-Optimal Exploration in Linear Dynamical Systems," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- [26] S. Dean, H. Mania, N. Matni, B. Recht, and S. Tu, "On the Sample Complexity of the Linear Quadratic Regulator," *Foundations of Computational Mathematics*, vol. 20, no. 4, pp. 633–679, 2020.
- [27] J. Kiefer and J. Wolfowitz, "The Equivalence of Two Extremum Problems," *Canadian Journal of Mathematics*, vol. 12, pp. 363–366, 1960.
- [28] F. Pukelsheim, *Optimal Design of Experiments*. SIAM Classics in Applied Mathematics, 2006.
- [29] R. D. Gill and B. Y. Levit, "Applications of the van Trees Inequality: A Bayesian Cramér–Rao Bound," *Bernoulli*, vol. 1, no. 1–2, pp. 59–79, 1995.
- [30] M. Simchowitz and D. J. Foster, "Naive Exploration is Optimal for Online LQR," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- [31] T. L. Lai and H. Robbins, "Adaptive Design and Stochastic Approximation," *The Annals of Statistics*, vol. 7, no. 6, pp. 1196–1221, 1979.